Inverting the path signature from its unitary development

Workshop on Mathematics of Data Streams: Signatures, Neural Differential Equations, and Diffusion Models

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DataSig

A rough path between mathematics and data science

The Alan Turing Institute

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Engineering and Physical Sciences Research Council

Outline

1. Introduction and motivation
2. Preliminaries
3. Main theorem and proof
4. A novel metric on probability measures on the path space
5. Numerical application: hypothesis testing
6. Conclusion

Motivation and objectives

Motivation

- A fundamental question in rough path theory: determine the signature from its Cartan development into unitary matrix groups, *a.k.a. unitary development*. [Chevyrev and Lyons, 2016]
- A more general problem: *Generalised moment problem on the (infinite-dimensional) tensor algebra*

  Determine a tensor $X \in \bigoplus_{n=0}^{\infty} \left(\mathbb{R}^d\right)^{\otimes n}$ *whose radius of convergence is non-zero from its moment generating function under unitary group.*

Our work

- Provide constructive proof for solving the generalized moment problem.
- A new metric on distributions on the path space and its applications in hypothesis testing.
- Working paper: Determination of the signature from the unitary development
- Joint work with Zijiu Lyu (Oxford), Siran Li (Shanghai Jiao Tong Uni.), Jiajie Tao (UCL).

Tensor series

Definition (Tensor algebra space)

The space of formal series of tensor of $\mathbb{R}^d$, denoted by $\mathcal{T}(\mathbb{R}^d)$, is defined by $\bigoplus_{n\geq 0}(\mathbb{R}^d)^{\otimes n}$. It is endowed with an elementwise addition and a tensor product.

- An alphabet $\mathcal{A} \equiv \{1, \dots, d\}$; any element in $\mathcal{A}$ is called a letter.
- Word:

  $$\mathbf{Word}(n) := \left\{ W \ : \ W \text{ is a word of length } n \text{ from the alphabet } \{1, \dots, d\} \right\}.$$

- Let $(e_i)_{i\in\mathcal{A}}$ be the basis of $\mathbb{R}^d$. Then a generic tensor $X \in \mathcal{T}\left(\mathbb{R}^d\right)$ can be expressed as the (formal) sum:

  $$X = \sum_{n=0}^{\infty} \sum_{W\in\mathbf{Word}(n)} X^W e_W \tag{1}$$

  where, for each $W = \left(w_1, \dots, w_n\right)$ as above, $X^W \in \mathbb{R}$ is a real coefficient and $e_W \equiv e_{w_1} \otimes \cdots \otimes e_{w_n}$.

Moment generating function of a tensor series

- Let $\mathfrak{gl}(k,\mathbb{C}) \cong \mathbb{C}^{k\times k}$ be the algebra of $k \times k$ complex matrices.
- $\mathfrak{su}(k) := \{M \in \mathfrak{gl}(k,\mathbb{C}) | M + M^* = 0\}$ - the Lie algebra of (special) unitary group.

Definition (Canonical extension of $M \in \mathscr{L}(\mathbb{R}^d, \mathfrak{gl}(k,\mathbb{C}))$)

Let $M \in \mathscr{L}(\mathbb{R}^d, \mathfrak{gl}(k,\mathbb{C}))$. $\widetilde{M} \in \mathscr{L}(\mathscr{T}\left(\mathbb{R}^d\right), \mathfrak{gl}(k,\mathbb{C}))$ is the canonical extension of M defined as follows:

$\widetilde{M}$ is the algebra homomorphism determined by $\widetilde{M}(1) = I_k$ and that

$$\widetilde{M}\left(e_{i_1} \otimes \cdots \otimes e_{i_\ell}\right) := M\left(e_{i_1}\right) \bullet \cdots \bullet M\left(e_{i_\ell}\right) \text{ for any } \ell \in \mathbb{N} \text{ and } i_1, \ldots, i_\ell \in \mathscr{A}. \tag{2}$$

Definition (Moment generating function of $X \in \mathcal{T}\left(\mathbb{R}^d\right)$)

Given a tensor $X \in \mathcal{T}\left(\mathbb{R}^d\right)$, the generating function of X is the mapping

$$\Phi_X : \bigcup_{k\in\mathbb{N}} \mathscr{L}(\mathbb{R}^d; \mathfrak{su}(k)) \longrightarrow \bigcup_{k\in\mathbb{N}} \mathfrak{gl}(k,\mathbb{C}); \Phi_X(M) := \widetilde{M}(X). \tag{3}$$

Remark

- Let $d = 1$ and $k = 1$. Then $X = (x_0, x_1, x_2, \cdots) \in \mathcal{T}\left(\mathbb{R}^d\right)$. The generating function of X is given by

$$\Phi_X(M_\lambda) = \sum_{n=0}^{\infty} x_n \lambda^n \tag{4}$$

  where $M_\lambda : x \mapsto \lambda x$ for $\lambda \in i\mathbb{R}$.
- To recover x_n, one can calculate x_n by differentiating Φ_X, i.e,

$$\frac{1}{n!}\frac{d^n}{d\lambda^n}\Phi_X(M_\lambda)|_{\lambda=0} = x_n \tag{5}$$

Path signature

Definition (Path signature)

Consider a path $\gamma : [0,T] \to \mathbb{R}^d$ of bounded p-variation for some $p \geq 1$. Suppose that the following integration is well-defined. The *signature of* γ, taking values in $\mathscr{T}\left(\mathbb{R}^d\right) := \bigoplus_{n=0}^{\infty} \left(\mathbb{R}^d\right)^{\otimes n}$ is defined as a sequence of iterated integrals:

$$S(\gamma)_{[0,T]} := \left(1, \gamma^{(1)}_{[0,T]}, \dots, \gamma^{(n)}_{[0,T]}, \dots\right) \tag{6}$$

where

$$\gamma^{(n)}_{[0,T]} := \int_{0<t_1<t_2<\ldots<t_n<T} \mathrm{d}\gamma_{t_1} \otimes \cdots \otimes \mathrm{d}\gamma_{t_n} \in \left(\mathbb{R}^d\right)^{\otimes n}, \tag{7}$$

with the convention $\left(\mathbb{R}^d\right)^{\otimes 0} \equiv \mathbb{R}$.

The development of a path

- G: matrix Lie group and $\mathfrak{g}$: its Lie algebra;
- Lie algebra examples:
  - $\mathfrak{su}(k) = \{M \in \mathfrak{gl}(k, \mathbb{C}) | M + M^* = 0\}$ - the Lie algebra of (special) unitary group.;
  - $\mathfrak{so}(k) = \{M \in \mathbb{R}^{k \times k} | M + M^T = 0\}$- the Lie algebra of special orthogonal group.

Definition (Development of the path)

Let $M \in \mathscr{L}\left(\mathbb{R}^d, \mathfrak{g}\right)$. The development of a path $\gamma : [0, T] \to \mathbb{R}^d$ under M is $\mathscr{D}_M(\gamma) = Y_T \in G$ such that Y_t is the solution to the following differential equation:

$$dY_t = Y_t \cdot M(d\gamma_t), Y_0 = \mathrm{Id} \tag{8}$$

where Id is the identity of G, and $\mathscr{L}\left(\mathbb{R}^d, \mathfrak{g}\right)$ is the vector space of linear transforms from $\mathbb{R}^d$ to $\mathfrak{g}$. We call the map $\mathscr{D}_\bullet(\gamma) : \mathscr{L}\left(\mathbb{R}^d, \mathfrak{g}\right) \to G$: $M \mapsto \mathscr{D}_M(\gamma)$ the development of γ under G.

Path signature v.s. development

Lemma (Connection with signature)

Let $X \in \mathscr{V}_1([0,T],\mathbb{R}^d)$ *and let* $M \in L(\mathbb{R}^d,\mathfrak{g})$. *Then* $\mathscr{D}_M(X) = \Phi_{S(X)}(M)$.

Underlying space	$x \in \mathbb{R}$	$X := (X_t)_{t\in[0,T]}$
Monominals	$(1, x, x^2, \cdots)$	$S(X)$
Moment-generating function	$\lambda \mapsto \exp(\lambda x)$	$M \mapsto \mathscr{D}_M(X)$.

	Signature	Development
Multiplicative property	Y	Y
Time-invariance	Y	Y
Trainable	N	Y
Universality	Y	Y ($\mathfrak{u}$)
Characteristic property	Y	Y ($\mathfrak{u}$)

Table: Comparison between the signature and development.

Applications of rough path theory in ML

Machine learning applications

Signature-based methods

- Methods: Linear signature model [Levin et al., 2013], deep signature transform[Kidger et al., 2019], signature kernel [Király and Oberhauser, 2019, Salvi et al., 2021].
- Complimentary features of sequential data to boost ML performance: Online handwritten character recognition [Xie et al., 2017], financial data analysis [Lyons et al., 2014], sepsis detection [Morrill et al., 2019], etc.
- Generative models for synthetic time series generation: e.g., [Liao et al.,].

Path development-based methods

- Supervised learning on sequential data: [Lou et al., 2022]; action recognition [Jiang et al., 2024].
- Generative models for synthetic time series generation: PCF-GAN [Lou et al., 2024]
- Lévy area generation for high-order SDE simulation [Jelinčič et al., 2023].

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Multiplicity and position functions

For each word $W \in \mathbf{Word}(n)$ and each letter of it, we define the multiplicity and position functions:

1. For $i \in \{1, \dots, d\}$, we set $r^W(i) :=$ the multiplicity of i in W, *i.e.*, the number of times that i appears in W.
2. For $i \in \{1, \dots, d\}$ and $t \in \{1, \dots, r^W(i)\}$, we set $p^W(i|t) :=$ the position that i appears in W for the tth time. We put $p^W(i|t) = \varnothing$ if $r^W(i) = 0$.

Example

Consider the case $d = 4$, $n = 5$, and $W = (12221)$. Then

$$r^W(1) = 2, \quad r^W(2) = 3, \quad r^W(3) = r^W(4) = 0;$$

as well as

$$p^W(1|1) = 1, \quad p^W(1|2) = 5, \quad p^W(2|1) = 2, \quad p^W(2|2) = 3, \quad p^W(2|3) = 4,$$
$$\text{and} \quad p^W(i|t) = \varnothing \quad \text{for all the other } i \in \{1,2,3,4\} \text{ and } t \in \{1, \dots, r^W(i)\}.$$

Matrix

- We use the symbol E_b^a $(a, b \in \{1, \dots, k\})$ to denote the *elementary matrices*:

$$E_b^a \in \mathfrak{gl}(k, \mathbb{C}) \quad \text{whose } (a, b)\text{-entry is 1 and all the other entries are 0.}$$

- Note that

$$\left\{ E_b^a - E_a^b \,:\, 1 \leq a < b \leq k \right\} \text{ is a natural basis for } \mathfrak{so}(k, \mathbb{R}).$$

- The following identity will be used repetitively:

$$E_b^a \bullet E_q^p = \delta_b^p E_q^a \qquad \text{for any } a, b, p, q \in \{1, \dots, k\}. \tag{9}$$

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Theorem

Given a mapping

$$\Phi : \bigcup_{k=0}^{\infty} \mathscr{L}\left(\mathbb{R}^d, \mathfrak{su}(k)\right) \to \bigcup_{k=0}^{\infty} \mathfrak{gl}(k, \mathbb{C})$$

graded by $k \in \mathbb{N}$. *Suppose that* Φ *is the generating function of some* $X \in \widehat{\mathscr{E}}$, *i.e., a tensor with the non-zero radius of convergence. One can determine* X *explicitly by the following procedure.*
(i) Write

$$X = \sum X^W e_{w_1} \otimes \cdots \otimes e_{w_n}, \qquad W \equiv \left(w_1, \ldots, w_n\right),$$

where $\{e_1, \ldots, e_d\}$ *is the Cartesian basis for* $\mathbb{R}^d$ *and the summation is taken over all possible* $n \in \mathbb{N}$ *and ordered* n*-tuples* $W \in \{1, \ldots, d\}^n$. *It suffices to determine* X^W *for each* W.
(ii) To this end, fix any $k \geq n + 1$ *and take the linear mappings*

$$\mathscr{M}[W]_1, \ldots, \mathscr{M}[W]_d \in \mathscr{L}\left(\mathbb{R}^d, \mathfrak{so}(k, \mathbb{R})\right)$$

defined for each $i, j \in \{1, \ldots, d\}$ *as follows:*

Theorem (Continued)

$$\mathscr{M}[W]_i(e_j) := \begin{cases} \sum_t' \left(E^{p^W(i|t)}_{p^W(i|t)+1} - E^{p^W(i|t)+1}_{p^W(i|t)} \right) & \text{if } i = j \in \{w_1, \ldots, w_n\}, \\ 0 \qquad \text{if } i \notin \{w_1, \ldots, w_n\} \text{ or } i \neq j. \end{cases}$$

(iii) Form the linear combination, $\mathscr{M}[W]_{(\theta)} = \sum_{i=1}^{d} \theta_i \mathscr{M}[W]_i$*, where* $\theta = (\theta_1, \ldots, \theta_d)^\top \in \mathbb{R}^d$ *are hyperparameters. Then*

$$X^W = \frac{1}{\mathscr{C}_W} \times \left\{ \text{the } (1, n+1)\text{-entry of the matrix } \frac{\partial^n \Phi\left(\mathscr{M}[W]_{(\theta)}\right)}{\partial\theta_{w_1} \cdots \partial\theta_{w_n}}\bigg|_{\theta=0} \right\}, \quad (10)$$

The combinatorial constant $\mathscr{C}_W$ *equals the product over each* $i \in \{1, \ldots, d\}$ *of the factorial of the number of times that* i *appears in* W*.*

Example

Example (Calculation of $\mathscr{M}[W]_{(\theta)}$)

Suppose $X \in \hat{E}(\mathbb{R}^2)$ and we aim to invert X_W where $W = (121)$.

- We choose $n = 4$ and define the linear maps $\{\mathscr{M}[W]_i\}_{i=1,2}$ as follows:

$$\mathscr{M}[W]_1(e_1) = E_2^1 - E_1^2 + E_4^3 - E_3^4 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad \mathscr{M}[W]_1(e_2) := 0, \tag{11}$$

$$\mathscr{M}[W]_2(e_1) := 0, \quad \mathscr{M}[W]_2(e_2) = E_3^2 - E_2^3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{12}$$

- $\mathscr{M}[W]_{(\theta)} = \sum_{i=1}^{d} \mathscr{M}[W]_i \theta_i \Rightarrow \Phi(\mathscr{M}[W]_{(\theta)})$.

Key gradients of the proof: (I)

Define for a hyperparameter $\theta = (\theta_1, \dots, \theta_d) \in \mathbb{R}^d$ the matrix

$$M_{(\theta)} := \sum_{i=1}^{d} \theta_i M_i, \tag{13}$$

where $M_1, \dots, M_d$ are arbitrary linear maps in $\mathscr{L}\left(\mathbb{R}^d, \mathfrak{su}(k)\right)$.

Key observation (I)

$$\frac{1}{\mathscr{C}_W} \frac{\partial^{|W|} \Phi_X(M_{(\theta)})}{\partial \theta^W}\bigg|_{\theta=0}$$

$$= \sum_{\overline{W} \in \mathbf{Word}(|W|)} X^{\overline{W}} \left\{ \sum_{\{I \in \mathbf{Word}(|W|):\, I \sim W\}} M_{i_1}\left(e_{\overline{w}_1}\right) \bullet \cdots \bullet M_{i_{|J|}}\left(e_{\overline{w}_{|J|}}\right) \right\}, \tag{14}$$

where $\overline{W} = \left(\overline{w}_1, \dots, \overline{w}_{|W|}\right)$, $I = \left(i_1, \dots, i_{|W|}\right)$, and $\mathscr{C}_W$ is the combinatorial constant $\mathscr{C}_W := \prod_{\alpha=1}^{|W|} \left[r^W\left(w_\alpha\right)!\right] = \prod_{i=1}^{d} \left[r^W(i)!\right]$.

Sketch of proof

The generating function of X acting on $M_{(\theta)}$ can be computed as follows:

$$\Phi_X(M_{(\theta)}) := \widetilde{\sum_{i=1}^{d} \theta_i M_i} \left(\sum_{n=0}^{\infty} \sum_{\overline{W} \in \mathbf{Word}(n)} X^{\overline{W}} e_{\overline{W}} \right)$$

$$= \sum_{n=0}^{\infty} \sum_{\overline{W}=(\overline{w}_1,\dots,\overline{w}_n)\in \mathbf{Word}(n)} X^{\overline{W}} \left\{ \sum_{i_1=1}^{d} \theta_{i_1} M_{i_1}\left(e_{\overline{w}_1}\right) \right\} \bullet \cdots \bullet \left\{ \sum_{i_n=1}^{d} \theta_{i_n} M_{i_n}\left(e_{\overline{w}_n}\right) \right\}$$

$$= \sum_{n=0}^{\infty} \sum_{\overline{W}\in \mathbf{Word}(n)} X^{\overline{W}} \cdot \left\{ \sum_{I=(i_1,\dots,i_n)\in \mathbf{Word}(n)} \left[\theta_{i_1} \cdots \theta_{i_n}\right] M_{i_1}\left(e_{\overline{w}_1}\right) \bullet \cdots \bullet M_{i_n}\left(e_{\overline{w}_n}\right) \right\}. \tag{15}$$

Moreover, it holds

$$\frac{d^{|W|}}{d\theta^W} \left[\theta_{i_1} \cdots \theta_{i_n}\right]|_{\theta=0} = \mathscr{C}_I I_{I\sim W}. \tag{16}$$

(II) Construction of the paramerizated linear map M_θ

All we need to do is to evaluate the $(1, n+1)$-entry of

$$\left\langle\left\langle I, \overline{W}\right\rangle\right\rangle_{\mathscr{M}[W]} = \mathscr{M}[W]_{i_1}\left(e_{\overline{w}_1}\right) \bullet \cdots \bullet \mathscr{M}[W]_{i_n}\left(e_{\overline{w}_n}\right);$$

Only non-zero term occurs when $I = \overline{W} = W$.

Case 1: $\overline{W} \neq I$. There exists some $\ell \in \{1, \ldots, n\}$ such that $i_\ell \neq \overline{w}_\ell$. Thus $\mathscr{M}[W]_{i_\ell}\left(e_{\overline{w}_\ell}\right) = 0$ and hence $\left\langle\left\langle I, \overline{W}\right\rangle\right\rangle_{\mathscr{M}[W]}$ equals the zero matrix.

Case 2: $I = \overline{W} \sim W$. In this case, we are interested in

$$\langle\langle I, I\rangle\rangle_{\mathscr{M}[W]} = \mathscr{M}[W]_{i_1}\left(e_{i_1}\right) \bullet \cdots \bullet \mathscr{M}[W]_{i_n}\left(e_{i_n}\right), \tag{17}$$

$$\Rightarrow \langle\langle I, I\rangle\rangle_{\mathscr{M}[W]} = \text{ linear combination of matrices of the form } E^{a_1}_{b_1} \bullet \cdots \bullet E^{a_n}_{b_n}, \tag{18}$$

where either $a_i + 1 = b_i$ or $a_i = b_i + 1$ for each $i \in \{1, \ldots, n\}$.

Proof

Thanks to the identity (9) for elementary matrices, we have

$$E^{a_1}_{b_1} \bullet \cdots \bullet E^{a_n}_{b_n} = \delta^{a_2}_{b_1} \cdots \delta^{a_n}_{b_{n-1}} E^{a_1}_{b_n}.$$

The only non-zero term contributed to the $(1, n+1)$-entry of $\langle\langle I, I\rangle\rangle_{\mathscr{M}[W]}$ is given by

$$a_1 = 1, \qquad b_n = n+1, \qquad \text{and } a_{\ell+1} = b_\ell \text{ for each } \ell \in \{1, \dots, n-1\}.$$

Then by the triangle inequality and $|a_i - b_i| = 1$, it follows that

$$\begin{aligned} n &= b_n - a_1 \\ &\leq |b_n - a_n| + |a_n - b_{n-1}| + |b_{n-1} - a_{n-1}| + |a_{n-1} - b_{n-2}| + \dots + |a_2 - b_1| + |b_1 - a_1| \\ &= \sum_{i=1}^{n} |b_i - a_i| = n, \end{aligned}$$

so equalities must hold everywhere. But this is only possible when

$$(a_1, \dots, a_n) = (1, \dots, n-1) \qquad \text{and} \qquad (b_1, \dots, b_n) = (2, \dots, n).$$

Example

Example

- Recall the previous example. $W = 121$.

$$M_1 := \mathscr{M}[W]_1(e_1) = E_2^1 - E_1^2 + E_4^3 - E_3^4 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \tag{19}$$

$$M_2 := \mathscr{M}[W]_2(e_2) = E_3^2 - E_2^3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{20}$$

- We compute:

$$M_1M_2M_1 = \begin{pmatrix} 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad M_1M_1M_2 = M_2M_1M_1 = \begin{pmatrix} 0 & 0 & 0 & \boxed{0} \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

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Restricted unitary development feature

Corollary (Seperation of points)

For any $X \in \widehat{\mathscr{E}}$ *with* $\pi_n(X) \neq 0$ *and any* $m \geq n+1$, *we can find* $M \in \mathscr{L}\left(\mathbb{R}^d, \mathfrak{t}(k)\right)$ *such that* $\Phi_X(M) \neq 0$. *Here* Φ_X *is the generating function of* X *as in Definition 3, and* $\mathfrak{t}(m, \mathbb{R})$ *is the space of tridiagonal antisymmetric* $m \times m$ *matrices:*

$$\mathfrak{t}(m) :=$$

$$\left\{ M \in \mathfrak{so}(m, \mathbb{R}) : \text{ the } (i,j)\text{-entry of } M \text{ is possibly nonzero only when } |i-j| = 1 \right\}.$$

Definition (Restricted unitary development feature)

Let $\gamma \in \mathscr{V}_p([0,T], E)$ and $m \in \mathbb{N}$. The restricted unitary development feature of γ of order m is defined as $\mathscr{D}_\gamma$ whose domain is restricted to $L(\mathbb{R}^d, \mathfrak{t}((m))$. Likewise, The restricted unitary feature of γ of all orders is defined as $\mathscr{D}_\gamma$ whose domain is given by $\oplus_{n \geq 1} L(\mathbb{R}^d, \mathfrak{t}((m))$.

RPCFD

- We restrict the path space to be the space of time-augmented paths, i.e.,

$$\tilde{\mathscr{V}}_p([0,T],E) := \{\tilde{X} | \tilde{X}(t) = (t, X(t)), \forall t \in [0,T]\}_{X \in V_p([0,T],\mathbb{R}^d)}.$$

- Let $\mathscr{P}_0\left(\tilde{\mathscr{V}}_1([0,T],\mathbb{R}^d)\right)$ denote the space of probability measures on $\tilde{\mathscr{V}}_1([0,T],\mathbb{R}^d)$, under which the corresponding expected signature has infinite radius of convergence.

Definition (RPCFD)

Suppose μ and ν are two probability measures in the space $\mathscr{P}_0\left(\tilde{\mathscr{V}}_1([0,T],\mathbb{R}^d)\right)$. Let $m \geq 1$ and denote by $\mathscr{M}$ any the probability measure on $\mathscr{L}(\mathbb{R}^d, \mathfrak{t}(m))$ with a full support. Then, the Restricted Path Characteristic Function Distance (RPCFD) between μ and ν w.r.t. $\mathscr{M}$ is defined by:

$$\mathrm{RPCFD}^2_{\mathscr{M}}(\mu,\nu) := \mathbb{E}_{M\sim\mathscr{M}}\left[\left|\left|\mathbb{ED}_M(\mu) - \mathbb{ED}_M(\nu)\right|\right|^2_{HS}\right], \tag{21}$$

where the expected development $\mathbb{ED}_M(\mu) := \mathbb{E}_{X\sim\mu}[\mathscr{D}_M(X)]$ and $\mathbb{ED}_M(\mu) := \mathbb{E}_{Y\sim\nu}[\mathscr{D}_M(Y)]$.

Remark

- The only difference between Characteristic function distance (PCFD) proposed in [Lou et al., 2024] and RPCFD lies in the range of the linear map M (i.e., $\mathfrak{su}(m)$ vs $\mathfrak{t}(m, \mathbb{R})$).
- Similar to PCFD, RPCFD enjoys uniform boundedness and differentiability w.r.t model parameters due to the fact that $\mathfrak{t}(m, \mathbb{R})$ is a proper subset of $\mathfrak{u}(m)$. The separation of points property of RPCFD is a direct consequence of the characteristic property of restricted unitary development feature.
- Note that $\dim(\mathfrak{t}(m, \mathbb{R})) = m - 1$ whereas $dim(\mathfrak{u}(m)) = m^2$. It means RPCFD offers a significant dimension reduction benefit over PCFD while retaining the characteristic property.

Trainable discriminator based on RPCFD

- To enhance the discriminative power of RPCFD on two given path distributions $\boldsymbol{X}$ and $\boldsymbol{Y}$, we propose the following trainable distance by

$$\sup_{\mathscr{M}\in\mathscr{P}(\mathscr{L}(\mathbb{R}^d,\mathbf{t}(m,\mathbb{R})))} \mathrm{RPCFD}^2_{\mathscr{M}}(\mu,\nu), \tag{22}$$

  where the supremum is taken over for all possible probability measures on $\mathscr{L}(\mathbb{R}^d,\mathbf{t}(m,\mathbb{R}))$ for given $m\geq 1$, denoted by $\mathscr{P}(\mathscr{L}(\mathbb{R}^d,\mathbf{t}(m,\mathbb{R})))$.

- Numerically, we approximate $\mathscr{M}$ by the empirical measure $\mathscr{K}_k$, i.e., there exists a finite $K\geq 1$, such that

$$\mathscr{M}_k := \frac{1}{K}\sum_{i=1}^{K}\delta_{M_i},$$

  where $M_i\in\mathscr{L}(\mathbb{R}^d,\mathbf{t}(m,\mathbb{R}))$ for $i\in\{1,\cdots,K\}$, which can be trained from data.

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Two-sample test

Two-sample test on stochastic processes

Let X and Y denote two stochastic processes. Given $\mathbf{X} := (X_1, \cdots, X_m)$ and $\mathbf{Y} := (Y_1, \cdots, Y_n)$ where $X_i(Y_i) \sim_{iid} X(Y)$. The two-sample is a statistical method to determine whether there is sufficient evidence to accept or reject H_0:

$$H_0 : \mathbb{P}_X = \mathbb{P}_Y \text{ (null hypothesis)},$$
$$H_1 : \mathbb{P}_X \neq \mathbb{P}_Y \text{ (alternative hypothesis)},$$

where $\mathbb{P}_X$ and $\mathbb{P}_Y$ denote the distributions of X and Y, respectively.

Permutation test

Denoted by $\mathbf{X} = \{X_1, X_2, \cdots, X_m\}$ and $\mathbf{Y} = \{Y_1, Y_2, \cdots, Y_n\}$ two sets of i.i.d. samples drawn from $\mathbb{P}_X$ and $\mathbb{P}_Y$, respectively. The permutation test is conducted in the following way. Given the statistic $T(\mathbf{X}, \mathbf{Y})$, we can apply all possible the permutations $\tau \in \sigma(m+n)$ to $\mathbf{Z} := (\mathbf{X}, \mathbf{Y})$ and obtain the set $\mathcal{T}$ via

$$\mathcal{T} := \left\{ T(\mathbf{Z}_{\tau(1):\tau(m)}, \mathbf{Z}_{\tau(m+1):\tau(m+n)}) \mid \tau \in \sigma(m+n) \right\}.$$

Under H_0, one can uniformly sampling from $\mathcal{T}$ to approximate $T(\mathbf{X}, \mathbf{Y})$. Then, given the significance level α, the null hypothesis H_0 is rejected if $T(\mathbf{X}, \mathbf{Y})$ exceeds the $(1-\alpha)\%$ quantile of $\mathcal{T}$. See [Chevyrev and Oberhauser, 2022, §7].

Hypothesis testing for the law of stochastic processes

Example (Fractional Brownian motion)

Let us consider the 3-dimensional Brownian motion $\boldsymbol{B}_{[0,T]}$ and the fractional Brownian motion $\boldsymbol{B}^h_{[0,T]}$ with various Hurst parameter $h \in (0,1)$ and conduct two-sample hypothesis testing.

- Under the null hypothesis $\boldsymbol{H}_0$, we simulate the iid samples from $\mathbb{P}_X(\mathbb{P}_Y, \text{resp}) = \boldsymbol{B}^h_{[0,T]}$. Under the alternative hypothesis $\boldsymbol{H}_1$, we simulate the iid samples from $\mathbb{P}_X = \boldsymbol{B}_{[0,T]}(\mathbb{P}_Y = \boldsymbol{B}^h_{[0,T]}, \text{resp})$.
- Training set $\mathscr{D}_{\text{train}}$ is generated by iid samples under $\boldsymbol{H}_1$. We train the RPCFD/PCFD distance on $\mathscr{D}_{\text{train}}$.
- Test data $\mathscr{D}_{\text{test}}$ is the mixture of data generated by $\boldsymbol{H}_0$ and $\boldsymbol{H}_1$ with equal probability.
- We perform the permutation test to decide whether to accept or reject the null hypothesis $\boldsymbol{H}_0$ with the significance level $\alpha = 0.05$.
- We use the test power and Type 1 error to measure the performance of the trained RPCFD/PCFD. The higher power is better, and the lower Type 1 error is better.

PCFD vs RPCFD

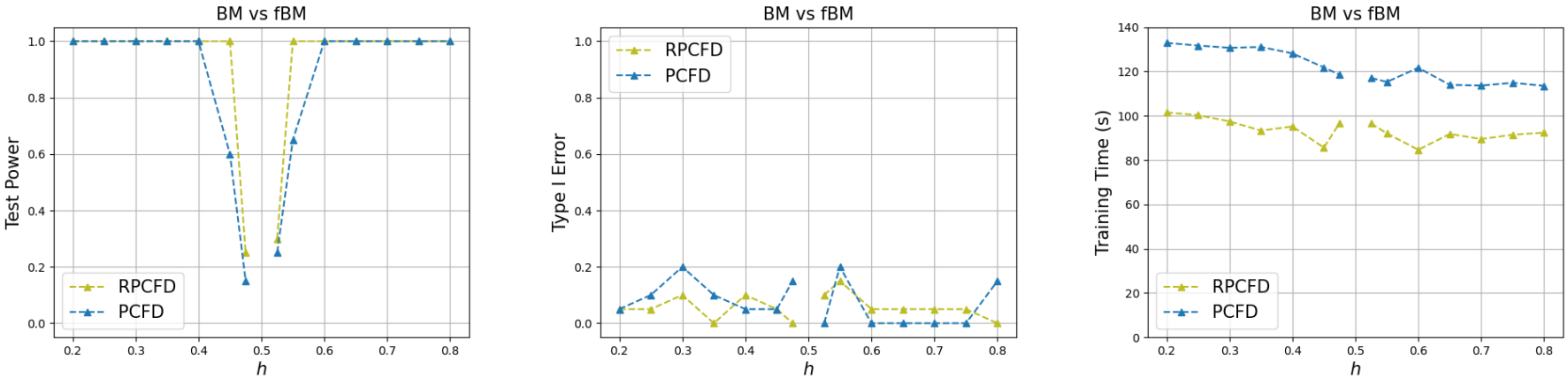


Figure: (Left) Test power and (Middle) of trained RPCFD and PCFD. (Right) Training time of RPCFD and PCFD over 500 iterations. Here fix $K = 8$ and $k = 5$.

Sensitivity analysis

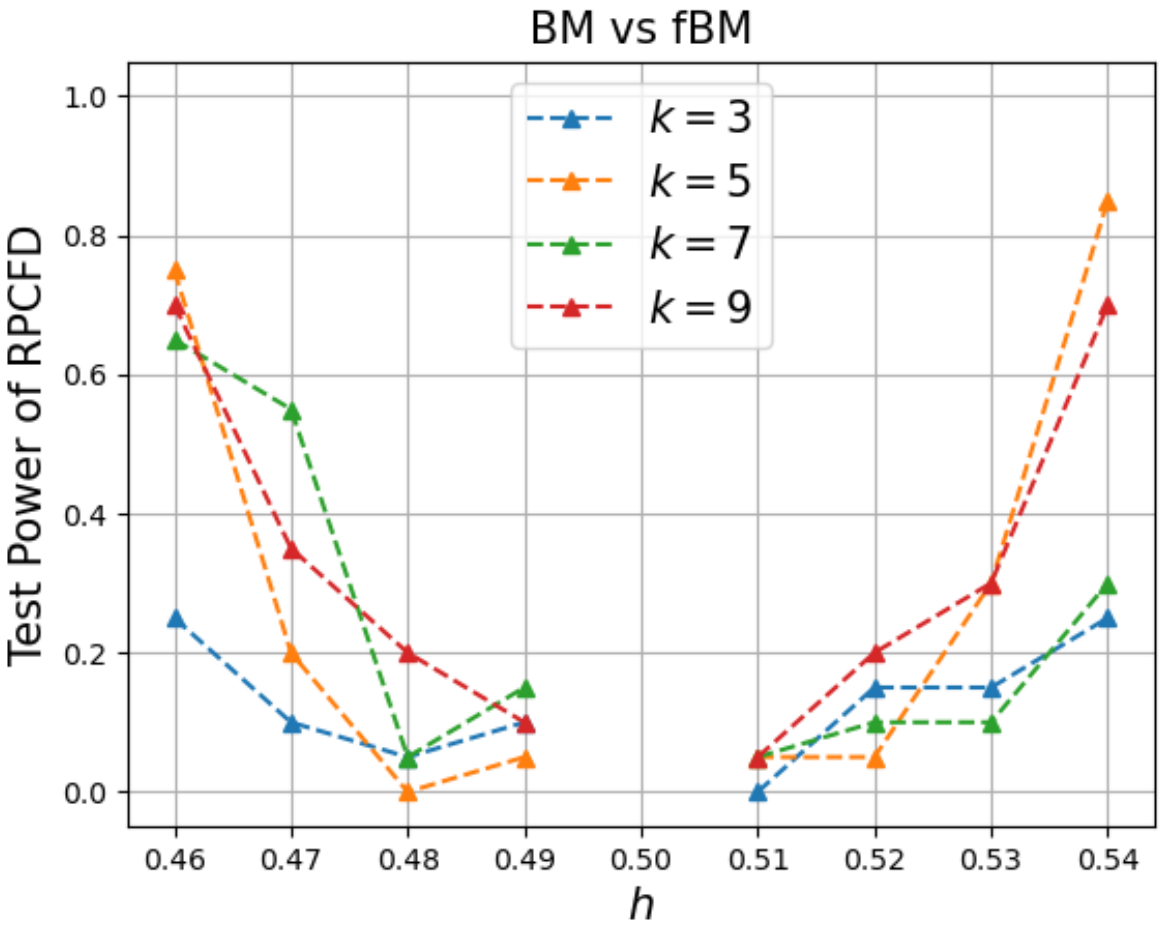


Figure: Sensitivity analysis of RPCFD. The y-axis and x-axis represent the mean power of the trained RPCFD and the Hurst parameter h, respectively. Here fix $K = 8$.

Comparison with Signature-MMD

	Developments			Signature MMDs	
h	RPCFD	OPCFD	PCFD	Linear	RBF
0.4	**1**	**1**	**1**	0.15	**1**
0.45	**1**	0.85	0.6	0	0.85
0.55	**1**	0.6	0.65	0.1	0.9
0.6	**1**	**1**	**1**	0.05	**1**

Table: Test power of the distances when $h \neq 0.5$.

	Developments			Signature MMDs	
h	RPCFD	OPCFD	PCFD	Linear	RBF
0.4	0.1	**0**	0.05	0.1	0.05
0.45	**0.05**	0.1	**0.05**	**0.05**	**0.05**
0.55	0.15	**0.05**	0.2	0.1	**0.05**
0.6	0.05	0.15	**0**	0.1	**0**

Table: Type I error of the distances.

For RPCFD, OPCFD and PCFD, fix $K = 8$ and $k = 5$. For RBF signature MMD, fix $2\sigma^2 = 0.1$.

Outline

1 Introduction and motivation

2 Preliminaries

3 Main theorem and proof

4 A novel metric on probability measures on the path space

5 Numerical application: hypothesis testing

6 Conclusion

Conclusion and Future work

Conclusion

- We provide a novel, constructive proof for solving the generalised moment problem on the tensor algebra.
- Generalise the classical results on the unique determination of signature by unitary development [Chevyrev and Lyons, 2016] - both by relaxing the condition on infinite radius of convergence (ROC) to nonzero ROC, and by extending to Cartan developments into a variety of matrix Lie groups.
- Propose a *novel and efficient metric (RPCFD) over the space of probability measures on the path space*, which features the sparsity of linear maps involved, hence bringing dimension reduction in hypothesis testing on the law of stochastic processes.

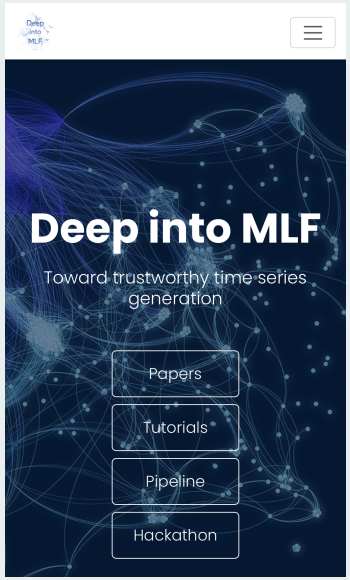
Future work

- Application of RPCF as the feature representation of time series to supervised learning.
- Application of RPCFD to generative models for synthetic time series generation.

Thanks for your attention!

More about our work

- UCL rough path and machine learning research group: https://www.ucl.ac.uk/rough-path-and-machine-learning-group
- GitHub for time series generation pipeline: https://github.com/DeepIntoStreams/Evaluation-of-Time-Series-Generative-Models.git.
- Online seminar series: Women in data science and mathematics (WINDSMATH): https://www.windsmath.com/
- FutureLearn course: An Introduction to Machine Learning in Quantitative Finance.



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